■ ESTIMATION STATION

How Many Do You Think? — Detailed Lesson Plan

■ Subject: Mathematics	■ Grade: 1–2	■ Duration: 45 minutes	■ PLANRS ACADEMY

■ Lesson Overview

This lesson introduces young learners to the concept of **estimation** — making smart, reasoned guesses about quantity without counting every item. Set in a vibrant Indian cultural context (sabzi mandi, laddoos, bindis, diyas), children follow Arjun on a trip to the mandi and engage in three rounds of guided estimation before practising independently at the Estimation Station.

■ Learning Objectives

Objective	Description
Know Estimation	Understand that estimation means using visual clues to make a reasonable guess — not a random one.
Make Smart Guesses	Apply known reference quantities (hands of 5, groups of 10) to estimate unfamiliar quantities.
Feel Proud	Build confidence in mathematical thinking by celebrating close estimates, not just exact answers.

■ Materials & Resources

Physical Props	Display & Recording
• Slide deck (16 slides, projected)	• Flowers / marigold garland (approx. 12 blooms visible)
• Closed dabba (box) with laddoos or counters inside	• Biscuit packet showing ~18 biscuits
• Small bowls of chana (or pebble substitutes)	• Diya shelf or diyas arranged for ~25 count

• Packet / display of bindis (or dot stickers)	• Student estimation journals / worksheets
• Pencil collection (25+ pencils bundled)	• Whiteboard / chalkboard for recording class guesses

■ Detailed Lesson Plan

Phase 1 — Hook & Introduction

■ 0–7 min | ■ Slides 1–3

Slide 1 — Title / Warm-Up

- Display the 'Estimation Station' title slide as children settle.
- Ask: *"Has anyone ever tried to guess how many sweets are in a jar?"*
- Take 3–4 quick responses; praise all answers with 'Wah!' or 'Shabash!'

Slide 2 — Learning Objectives

- Read out the three objectives together: **Know Estimation** → **Smart Guesses** → **Feel Proud!**
- Point to 'Use clues' — stress that estimation is NOT random guessing.
- Point to '10 fingers' — previewing the 10-frame anchor they'll use.

Slide 3 — The Mystery Dabba

- Show the actual closed dabba (or use slide image).
- Ask: *"How many laddoos do you think are inside? Show me on your fingers!"*
- Record 5–6 guesses on the board. Do NOT reveal the answer yet — build suspense!

Phase 2 — Direct Instruction

■ 7–18 min | ■ Slides 4–7

Slide 4 — Meet Arjun at the Mandi

- Introduce Arjun and his Amma going to the sabzi mandi to buy moong dal.
- Set the scene: *"The dal shop has SO many bags. Arjun needs to estimate!"*
- Discuss: when do we estimate in real life? Accept 2–3 ideas from students.

Slide 5 — What is Estimation?

- Teach the three-word definition using the visual: **USE CLUES** → **GET CLOSE** → **SMART GUESS!**
- Emphasise: estimation is *not* a wild guess — it uses what you can see.
- Use think-pair-share: *"Tell your partner one clue you could use to estimate."*

Slide 6 — The 3-Step Estimation Process

- **Step 1: LOOK** — observe the group carefully; note size, arrangement.
- **Step 2: THINK** — compare to something you know (5 fingers, a row of 10).
- **Step 3: GUESS** — say your estimate aloud with a reason.
- Practise chorally: *"LOOK… THINK… GUESS!"* (can add hand actions).

Slide 7 — Teacher Demo (Candy Jar)

- Hold up (or point to) a jar of colourful balls on screen.
- Model think-aloud: *"I can see about 10 in the front layer… there might be 4 more behind… so I estimate about 14!"*
- Count aloud to verify — celebrate if the estimate is within 5.

Phase 3 — Guided Practice

■ 18–33 min | ■ Slides 8–10

Slide 8 — Round 1: Chana (Chickpeas)

- Show 1 chana + 6 more = 7 chana total.
- Hint provided: *"Remember — 5 is one hand!"*
- Students write or show their guess on whiteboards / fingers.
- Reveal answer (7). Ask: *"Who was within 2? Within 5? Wah!"*

Slide 9 — Round 2: Bindis

- Show Group 1 (5 bindis) + Group 2 (8 bindis) = 13 bindis total.
- Hint: introduce the 10-frame visual (*"10 looks like two rows of 5"*).
- Ask children to hold up fingers showing their guess before you reveal.
- Reveal answer (13). Ask: *"Were you within 5? That's EXPERT level!"*

Slide 10 — Round 3: Pencils (Challenge!)

- **No hints given** — this is the challenge round!
- Flash the '? pencils' image briefly (2–3 seconds) then hide it.
- Students make quick estimates. Reveal reference (25 pencils).
- Celebrate estimates within 5 with 'Shabash!' badge acknowledgement.

Phase 4 — Independent Practice

■ 33–42 min | ■ Slide 11

Estimation Station — 3 Stations

- Divide class into 3 groups. Each group visits one station for ~3 minutes then rotates.
- ■ **Flower Stall** — estimate the marigolds: answer ~12 flowers.
- ■ **Biscuit Box** — estimate the biscuits: answer ~18 biscuits.
- ■ **Diya Shelf** — estimate the diyas: answer ~25 diyas.
- Students record their estimate in journals **before** checking the label.
- Teacher circulates, asking: *"What clue did you use?"*

Phase 5 — Wrap-Up & Assessment

■ 42–45 min | ■ Slides 12–16

Slide 12 — Close Enough is Great!

- Reinforce: if your guess is within 5 of the real number, you are an EXPERT estimator!
- Show the 'Wah! Shabash! Trophy' graphic — celebrate the class effort.

Slide 13 — Real Life Estimation

- Quick class discussion: *"Where do we estimate every day?"*
- Point to labels: Sabzi Mandi, Kitchen Dal, Diwali Diyas, Wedding Puris, Pookalam flowers.

- Ask: *"Has your Amma or Nani ever estimated at home?"*

Slide 14 — Recap Rhyme

- Chant together: **LOOK… THINK… GUESS!** with actions.
- Can repeat 2–3 times with increasing speed for energy.

Slide 15 — Exit Ticket

- Read the Riya scenario: she sees marbles in two groups of 10 → estimates ~20.
- Question: *"Is Riya A) Estimating or B) Just Guessing?"*
- Children show fingers: 1 for A, 2 for B. Answer: **A — she used clues and gave a reason!**

Slide 16 — Closing Celebration

- Celebrate: *"Wow — you did great today! Keep estimating!"*
- Optional homework: find one thing at home to estimate tonight and report back.

■ Differentiation Strategies

Support	Core	Extension
• Use physical objects that can be grouped into 5s. • Allow finger counting during estimation. • Pair with a stronger estimator during stations.	• Follow the 3-step process independently. • Record estimates in journals. • Explain one reason for each estimate.	• Estimate quantities over 30. • Write their estimate AND a range (e.g., 20–25). • Create their own estimation challenge for a friend.

■ Assessment & Success Criteria

Method	What to Look For	Success Indicator
Observation (Rounds 1–3)	Student verbalises a reason for their guess; references a known quantity (hand, group of 10).	Uses clue-based language: 'I think about 10…'
Station Journals	Written estimate is within ±5 of the actual count for at least 2 of 3 stations.	2/3 stations within ±5 = Estimation Expert ■
Exit Ticket (Slide 15)	Student correctly identifies Riya as 'estimating' and can explain she used clues.	Correct answer + 1 sentence reason.

■ Key Vocabulary

Term	Definition
Estimation	A smart, reasoned guess about quantity using visual clues — not an exact count.
Clue	A piece of information (size, grouping, reference number) used to guide a guess.
Reference Number	A known quantity used as a benchmark, e.g., 5 (one hand) or 10 (two hands / one row).
10-Frame	A visual tool showing two rows of 5 boxes, helping children anchor the idea of 10.
Range	A span of numbers that an estimate might fall within (e.g., 'between 10 and 15').

Shabash / Wah!	Hindi words of praise meaning 'Well done!' / 'Wow!' — used to celebrate good thinking.

■ Teacher Tips & Cultural Notes

■ Cultural Relevance	*The lesson uses familiar Indian objects (laddoos, chana, bindis, diyas, pookalam flowers). Encourage children to share their own home examples — this validates their cultural knowledge.*
■ Language Support	*The Hindi vocabulary (dabba, mandi, amma, shabash, wah) is used naturally. If children are not familiar, briefly translate — this is also a language enrichment opportunity.*
■ Praise Strategy	*Always praise the reasoning, not just the accuracy. 'I love that you used your fingers as a clue!' is more powerful than 'Correct!'*
■ Misconception Alert	*Some students may think estimation means guessing randomly. Reinforce throughout: 'A good estimate always has a reason.'*
■ Pacing	*Round 3 (pencils, no hints) is intentionally harder. If children struggle, briefly model Step 1–2 again before asking for guesses.*
■ Extension Homework	*Ask families to join in: 'How many rotis did we make tonight? How many steps to the gate?' This builds home–school connection.*

■ Student Recording Sheet (Photocopy-Ready)

Name: ___________________________________ Class: ________ Date: _______________

Station / Object	My Estimate	Clue I Used	Actual Count	Within 5? ✓/✗
■ Flower Stall				
■ Biscuit Box				
■ Diya Shelf				
■ Chana Bowl (bonus)				
✏■ Pencil Jar (bonus)				

Reflection: My estimate was good because I used the clue:

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